ICC-1 Vascular surgery: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

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1. Principles of arterial and venous diseases
2. Aneurysms
3. Vascular presentations

PART 1: LECTURE

Acute arterial ischaemia and vascular trauma [089] (lectures and LGW)

Topic: Principles of arterial and venous diseases

Definition and causes

- Acute limb ischaemia (ALI) = sudden fall in limb perfusion threatening viability; acute if symptoms < 2 weeks, chronic if > 2 weeks. Time-sensitive.
- Causes: embolism, thrombosis, trauma. ~80% of emboli are cardiac; atrial fibrillation is the commonest cause (also recent MI mural thrombus, valve disease, endocarditis, myxoma). Stopping warfarin in AF raises embolic risk.
- Embolus: sudden, no prior claudication, embolic source. Thrombosis: on an atherosclerotic plaque, graft or aneurysm; prior claudication (acute-on-chronic).

Clinical picture

- 6 Ps: pain, pallor, pulselessness, paraesthesia, paralysis, poikilothermia. Pain = usual presenting symptom; pulselessness = defining feature; paralysis = most ominous (may be irreversible).
- Colour: pale, then cyanosed, then mottled; fixed mottling = late. Empty veins, cold limb with a level, slow capillary refill.
- Sensory loss order: light touch, vibration, proprioception, then deep pain/pressure (late). Intrinsic foot muscles weaken first.
- Pale, cold, pulseless foot = ischaemia even with weakness. Saddle embolus mimics paraplegia: no sensory level and absent pulses.
- Category III (irreversible): anaesthesia, paralysis/rigid muscles, no arterial or venous Doppler signal.

Investigations

- Little time for tests. Duplex: quick, operator dependent, misses aorta/iliacs. CTA: fast, shows site and cause. DSA: invasive, before thrombolysis. MRA: no radiation, slow.

Management

- Start IV heparin + analgesia + optimise underlying condition, then revascularise.
- Embolus, viable limb: minimal work-up (ECG, CXR, bloods), Fogarty balloon embolectomy via common femoral arteriotomy.
- Thrombosis: thrombolysis if focal lesion; bypass if extensive.

Complications

- Reperfusion injury: hyperkalaemia (arrhythmia), myoglobin causing acute tubular necrosis, ARDS, myocardial depression.
- Compartment syndrome after revascularisation: tense, pain on passive stretch, pulses may be present; urgent fasciotomy (calf, anterior tibial compartment).

Vascular trauma

- Blunt, penetrating, iatrogenic. Hard signs (absent pulses, active bleeding, expanding/pulsatile haematoma, bruit/thrill, distal ischaemia): operate now, no imaging.
- Soft signs (small stable haematoma, nerve injury, unexplained hypotension, past bleeding, proximity): measure ABI; ABI < 0.9 = arteriography.
- Knee dislocation with pale pulseless foot = popliteal artery injury.
- ABCDE, control bleeding; repair directly, by graft, endovascular, or ligation (damage control).

Source: Acute Arterial Ischaemia and Vascular Trauma lecture [089]; LGW Acute Limb Ischaemia

PART 1: LECTURE

Chronic arterial ischaemia [088] (lecture and LGWs)

Topic: Principles of arterial and venous diseases

Causes and sites

- Over 45: atherosclerosis. Under 45: presenile atherosclerosis in diabetics; in non-diabetics Buerger's disease, Raynaud's, arteritis.
- Aortoiliac: buttock, thigh and calf claudication; Leriche syndrome = impotence + claudication + buttock wasting (absent femoral pulses).
- SFA (adductor canal, commonest site): calf claudication only; profunda collaterals supply thigh. Claudication is one level below the block.
- Infrapopliteal (diabetes, Buerger's): poor collaterals, so limb-threatening; worst with aortoiliac disease (multilevel).

Clinical features

- Claudication: pain on exercise relieved by rest. Neurogenic pseudoclaudication: bilateral, paraesthesia, back pain, provoked by standing still, more downhill, normal pulses.
- Rest pain: burning forefoot pain at night, worse with leg horizontal, relieved by hanging leg down; opioid-resistant; leads to gangrene without reconstruction.
- Arterial ulcer: punched out, very painful, no granulation, lateral ankle/toe tips, absent pulses, shiny hairless skin.

Investigations

- ABI = higher ankle pressure / higher arm pressure. > 1.2 calcified (diabetes, unreliable); 1.0-1.2 normal; 0.5-0.9 claudication; 0.1-0.4 critical ischaemia (CLI: ABI ≤ 0.5).
- Doppler waveform: triphasic, then biphasic, then monophasic.
- Healing needs ankle > 70 mmHg / toe > 50 mmHg. Imaging (CTA, or MRA if kidney function poor) only when intervention planned.

Management

- Claudication is usually benign (5% amputation in 5 years); 50% improve with stopping smoking + exercise. Commonest death = heart attack/stroke.
- Best medical therapy: stop smoking, statin, antiplatelet, BP and diabetes control, exercise, foot (podology) care. Screen heart and carotids (duplex, ECG/echo).
- Intervene for lifestyle-limiting claudication, rest pain, tissue loss. Endovascular poor for long, heavily calcified, orifice or uncrossable lesions.
- Grafts: synthetic (Dacron, ePTFE) or biological (autologous vein).
- Compartment syndrome after arterial repair: fasciotomy.

Buerger's disease

- Young male (< 40) heavy smokers; small/medium distal vessels (tibial, plantar, radial); upper and lower limbs; Raynaud's, thrombophlebitis migrans.
- Angiography of all four limbs: distal segmental skip lesions, corkscrew collaterals.
- Stopping smoking arrests (does not reverse) it; sympathectomy for rest pain/ulcers; bypass rarely feasible; amputation for gangrene.

Source: Chronic Arterial Ischaemia lecture (Dr Hesham Amer) [088]; LGW Chronic Arterial Ischaemia; Chronic Ischaemia LGW

PART 1: LECTURE

Venous vascular diseases [087] (lecture and LGWs)

Topic: Principles of arterial and venous diseases

Venous return

- Helped by calf muscle pump, one-way valves, negative intrathoracic pressure. Veins: superficial, deep, perforators.
- Venous insufficiency = failure of valves, patency or calf pump.

Deep vein thrombosis

- Virchow's triad: endothelial damage, stasis, hypercoagulable blood. Risks: age, surgery (pelvis, hip, leg), cancer, immobility, pregnancy, pill, factor V Leiden, protein C/S or antithrombin III deficiency, antiphospholipid.
- Often silent; symptoms ~3rd-4th day after surgery: swelling, pain, tenderness. Whole-leg swelling = iliofemoral. Homans' sign obsolete.
- Wells < 2 (low): D-dimer; if positive, compression ultrasound. Wells > 2: ultrasound directly. Duplex is the investigation of choice.
- Treatment: LMWH + warfarin/NOAC together; stop heparin when INR 2-3.
- PE with anticoagulation contraindicated: IVC filter.
- Prophylaxis: early mobilisation, hydration, intra-operative calf compression, stockings, SC LMWH from before surgery while at risk.
- Complications: PE, post-thrombotic leg, venous ulcer, phlegmasia alba/cerulea dolens.

Varicose veins

- Dilated, elongated, tortuous veins; mostly primary; more in women and with prolonged standing. Secondary: pelvic mass, previous DVT.
- Medial thigh/calf = great saphenous; posterolateral calf = small saphenous. Examine standing; cough thrill at SFJ.
- Trendelenburg test: high thigh tourniquet keeps veins empty = SFJ incompetence; otherwise move lower.
- Duplex = investigation of choice (junctions, perforators, deep veins).
- Complications: pain, swelling, bleeding, venous eczema (itchy red patch), thrombophlebitis, lipodermatosclerosis, ulcer.
- Treatment: stockings (minor, elderly, pregnant, unfit); sclerotherapy (small/medium below knee); endovenous laser; surgery (> 4 mm, complications, failed compression) = SFJ ligation + stripping.

Venous ulcers

- ~85% of chronic leg ulcers; 15-30% also arterial disease (mixed ulcer).
- Venous: gaiter area above medial malleolus, sloping edges, pigmentation, lipodermatosclerosis, eczema. Arterial: punched out, toes/lateral malleolus, absent pulses. Neuropathic: under metatarsal heads, thick keratinised edge.
- Ix: duplex, ABPI, CBC, ESR/CRP, lipids, glucose.
- Treatment: mobilise, elevate, compression (key; safe only if ABPI > 0.8). After healing: class I-II stockings; vein surgery if no deep incompetence.

Source: Venous Vascular Diseases lecture [087]; LGW Venous Vascular Disease; Varicose Veins LGW; Venous Ulcers LGW

PART 1: LECTURE

Aneurysms [085] (lecture and LGW)

Topic: Aneurysms

Definition and types

- Aneurysm = permanent localised dilatation of an artery to 1.5 x its normal diameter; can occur anywhere. Commonest = abdominal aortic (AAA), ~95% infrarenal.
- Classified by aetiology (atherosclerotic, traumatic, congenital, inflammatory, dissecting), structure (true/false), shape (fusiform/saccular) and site.
- True = wall has all 3 arterial layers. False (pseudo) = haematoma communicating with the lumen through a partial tear; wall = adventitia + compressed tissue.
- Pseudoaneurysm: mostly iatrogenic after puncture (femoral commonest), post-surgical (anastomotic), or infected in IV drug users. Duplex to diagnose; small asymptomatic = conservative; > 2.5 cm = US-guided thrombin injection; or open repair.

AAA: presentation and investigation

- Most found incidentally on U/S; pulsatile mass, back pain, compression (vein, nerve, ureter, vertebrae). Smoking = key risk.
- Rupture: abdominal pain + back pain + hypotension; mortality > 80%. Can mimic pancreatitis, peritonitis, duodenal perforation, MI. Contained retroperitoneal rupture gives flank/groin pain, mistaken for ureteric colic.
- U/S = screening and surveillance; CTA/MRA = planning (CTA also shows thoracic aorta).
- SVS screening: all men ≥ 65; men ≥ 55 with family history; women ≥ 65 only with family history or smoking.

AAA: management

- Repair asymptomatic AAA at 5.5 cm (rupture risk > 5% surgical mortality). Open or EVAR depending on anatomy, age, comorbidity, surgeon, equipment, patient preference.
- EVAR: lower early mortality (1.2% vs 4.6%) but needs lifelong surveillance for migration, iliac limb occlusion, endoleak (flow into the sac).
- Complications of both: renal dysfunction, paraplegia, ischaemic colitis, graft infection; aorto-enteric fistula (rare, more after open) = GI bleeding.
- Ruptured AAA: ABCDE, permissive hypotension, urgent CTA, immediate surgery.
- Control smoking, lipids and BP to slow growth.

Visceral and peripheral aneurysms

- Splenic: commonest visceral; mostly women, linked to pregnancy. Treat if symptoms, growth, > 2.5 cm, or pregnancy (planned or current). Vaccinate as for splenectomy.
- Renal artery: 2nd commonest visceral; hypertension; repair if hard-to-control BP or > 2 cm.
- ~90% of peripheral aneurysms are femoral/popliteal; popliteal ~70%. Limb (not life) threatening: thrombosis, distal embolism (blue toe), vein/nerve compression.
- Prominent wide popliteal pulse = suspect aneurysm; 50% bilateral, check the aorta. Repair femoral > 2.5 cm, popliteal > 2 cm; exclusion + bypass (in situ saphenous vein preferred) or covered stent.

Source: Aneurysms lecture [SU.VAS.10.1]; LGW Aneurysms [SU.VAS.11.2]

PART 1: LECTURE

Vascular history taking and examination [104]

Topic: Vascular presentations

History

- Analyse symptoms: onset, provoking/relieving factors, quality, radiation, severity, timing. Claudication: distance, site (buttock, thigh, calf, foot), relief with rest. Rest pain: disturbs sleep, relieved by position.
- Buttock claudication + new erectile dysfunction = aorto-iliac occlusion (Leriche's syndrome).
- Pain after meals + weight loss = coeliac/SMA/IMA ischaemia. TIA, fleeting blindness, speech problems = carotid disease.
- Pack-years = packs (20 cigarettes) per day x years. Ask about past and passive smoking. Family history: AAA in 1st-degree relative, hypercoagulable states, early MI.
- Drug history: antiplatelets, anticoagulants, statins, OCP/HRT, antihypertensives.

Arterial examination

- General: radial pulse rate/rhythm, BP both arms, abdomen for visible/expansile aortic pulsation and bruit above the umbilicus.
- Colour: waxy white = acute severe; blue fixed mottling = acute irreversible; dark purple shiny = chronic ischaemia. Trophic changes: shiny skin, hair loss, brittle nails.
- Ischaemic ulcer: very painful, deep, punched out, toes or lateral ankle, grey granulation.
- CRT normal < 2 s; > 5 s = significant ischaemia.
- Pulses proximal to distal: femoral at mid-inguinal point (halfway ASIS to pubic symphysis); popliteal deep, patient prone, knee flexed; posterior tibial behind medial malleolus; dorsalis pedis lateral to EHL (absent in 10%). Doppler if not felt.
- Buerger's angle < 20 degrees = severe ischaemia; purple on dependency = positive Buerger's test.

Venous examination (standing)

- General: pulse, BP, neck veins, abdominal dilated veins. Veins crossing the groin = secondary varicose veins.
- Venous ulcer: large, shallow, sloping edges, medial ankle (gaiter), mild pain. Also eczema, haemosiderin, atrophie blanche, lipodermatosclerosis (inverted champagne bottle).
- Saphena varix: soft bluish swelling below and lateral to pubic tubercle, cough thrill, vanishes on lying.
- Hard, tender, cord-like vein = thrombophlebitis. Bruit over a varicosity that does not collapse on lying = AV malformation.

Special tests and duplex

- Trendelenburg: tourniquet at SFJ, stand. Fills from above on release = SFJ incompetence; fills from below with tourniquet on = incompetent perforators.
- Modified Perthes': tourniquet + walking; veins distend with severe pain = occluded deep system.
- Duplex before any VV treatment: source of reflux, suitability for ablation, deep vein function.

Source: Vascular history and examination LGW notes, checklists and VV notes [SU.VAS.01.2]

PART 1: LECTURE

Extracranial, renal and mesenteric ischaemia [080]

Topic: Vascular presentations

Carotid disease

- Stroke = focal (occasionally global) loss of cerebral function > 24 h with vascular cause; TIA < 24 h. 80% ischaemic, 20% haemorrhagic.
- Main cause of carotid territory stroke: thromboembolism from plaque at the ICA origin. Risks: HTN, DM, smoking, IHD, TIA, hyperlipidaemia.
- Symptoms: hemisensory/motor signs, dysphasia, amaurosis fugax (monocular blindness); FAST.
- Duplex = first line; CTA/MRA before surgery for arch and intracranial vessels; also bloods, echo, 24 h ECG tape.
- Management: best medical therapy + CEA or stenting; symptomatic = intervene ideally within 2 weeks (7-day stroke risk up to 8-10%).
- CEA complications: stroke 2-3%, bleeding/haematoma, cranial nerve injury (XII, X, IX).

Renal disease and dialysis access

- Atherosclerotic stenosis = commonest renal artery disease (1-5% of hypertensives); low perfusion activates RAAS. Treat medically +/- endovascular or surgical revascularisation.
- RRT: haemofiltration, peritoneal dialysis, haemodialysis, transplant.
- Vein mapping: artery > 2 mm, vein > 2.5 mm. Native AVF 6 months before need; graft 6-8 weeks before.
- Rule of 6s: flow ≥ 600 mL/min, < 6 mm deep, diameter ≥ 6 mm, straight length for 2 needles.
- Sites: radiocephalic, ulnobasilic, brachiocephalic, brachiobasilic transposition.
- Complications: stenosis, thrombosis, infection, failure to mature, steal (4 stages), high-output heart failure, aneurysm.

Mesenteric ischaemia

- Acute: embolism (usually SMA just beyond middle colic; limited bowel, better survival), thrombosis (at vessel origin; extensive bowel), NOMI, venous thrombosis.
- NOMI causes: low output (CHF, MI, sepsis, major surgery), vasopressors, ergotamines, cocaine, digitalis.
- Acute features: sudden severe pain out of proportion to signs, urge to defecate, fever, vomiting; raised WCC and lactate. Complications: sepsis, bowel death, death.
- CTA (or contrast CT) = investigation of choice. Treat: revascularisation (embolectomy, bypass, thrombolysis, angioplasty/stent) + resect dead bowel.
- Chronic: pain ~30 min after eating, lasting 1-3 h; food fear, weight loss; can become acute-on-chronic. CTA.
- Venous thrombosis: > 80% have a cause; hypercoagulable states (polycythaemia vera most common), tumour, infection, cirrhosis, trauma, pancreatitis.

Source: Extracranial, renal and mesenteric ischaemia [SU.VAS.15.1]

PART 1: LECTURE

Chronic limb swelling [084] (lecture and LGWs)

Topic: Vascular presentations

Causes and mechanisms

- Oedema mechanisms: increased permeability (cellulitis, angioedema, sepsis); increased capillary pressure (venous insufficiency, heart failure); low oncotic pressure (liver disease, nephrotic syndrome, protein-losing enteropathy, malabsorption); impaired lymph drainage (lymphoedema).
- Unilateral = local (DVT, cellulitis, filariasis); bilateral = systemic (heart, kidney, liver failure). Painful = DVT, cellulitis; painless = systemic, venous insufficiency, lymphoedema.
- Drugs: CCBs (~50% get oedema), NSAIDs, steroids.
- Myxoedema: mucopolysaccharides bind water = non-pitting boggy oedema, dry coarse skin; check TSH.
- Lipoedema: always bilateral, symmetrical, feet spared.
- Klippel-Trenaunay: lateral venous complex, capillary naevus, hypoplastic deep veins, limb lengthening.

Lymphoedema

- High-protein fluid from defective lymph drainage. Pitting at first; later hard and non-pitting. Stemmer's sign: cannot pinch skin at base of 2nd toe.
- Primary: congenita (infancy, males, bilateral; Milroy's = congenital + hereditary); praecox (commonest, females, after menarche, unilateral, to the knee); tarda (adult onset).
- Secondary: malignancy, surgery (axillary clearance), radiotherapy; filariasis (Wuchereria bancrofti) in developing countries. Exclude cancer recurrence in the axilla.
- Stages: 1 reversible, soft pitting, settles with elevation; 2 irreversible, hard; 3 elephantiasis with papillomas.
- Lymphoscintigraphy: tracer in first web space, gamma camera.
- Treatment: skin care, massage (manual lymph drainage), bandages, garments, pneumatic compression, exercise, complete decongestive therapy; surgery in severe cases. Antibiotics for infection.
- Arm care after axillary surgery: protect skin, no injections/blood tests/BP on that arm, avoid heat and tight items, lose weight.
- Lymphangiosarcoma: rare, years later, multiple subcutaneous nodules; distinguish from recurrent breast cancer.

Post-thrombotic syndrome

- After DVT: aching, heaviness, itching, swelling, secondary varicose veins, brown ankle pigmentation, ulcer. Iliofemoral obstruction: whole-leg swelling, abdominal wall collaterals, venous claudication.
- Secondary VV clues: previous DVT, irregular pattern, veins crossing the groin, ulceration.
- Duplex: deep veins, missed DVT, perforators; bloods for clotting problems.
- Treatment: mobilise, elevate, compression to heal ulcer, antibiotics if infected, then lifelong compression stockings.

Investigations

- CBC, urine and renal function, LFT, albumin, lipids, TSH, D-dimer (DVT); CXR, ECG, echo; abdominal U/S; venous duplex.

Source: Chronic limb swelling lecture [SU.VAS.12.1]; LGWs Chronic limb swelling, Lymphoedema, PTS [SU.VAS.13.2]

PART 1: LECTURE

Principles of limb ablation and supportive therapy [082]

Topic: Vascular presentations

Why and who

- PVD is the leading cause of lower limb amputation in the Western world; diabetics are 8-12 times more likely to need amputation.
- Causes: critical limb ischaemia, sepsis, trauma, neoplasm, congenital.
- Outcome is poor: mortality approaches 50% at 2 years in diabetics; half of diabetics with a major amputation lose the other leg within 5 years.
- CLI: revascularise first (angioplasty or bypass); amputate if this fails, or primarily by choice or comorbidity.

Diabetic foot and level choice

- Foot sepsis with palpable pulses: good prognosis with early surgery. Absent pulses: revascularise as early as possible.
- Digital gangrene: amputate digit + metatarsal head, leave wound open; infection is often more extensive than it looks, so warn of a higher level at consent.
- Level depends on rehab potential, vascular disease and likely healing; proximal levels need more walking energy. Check blood supply before toe amputation.

Operative principles

- Do not undermine flaps; tourniquet cuts blood loss without harming healing.
- Divide nerves cleanly, away from bone ends (prevents neuroma). Muscle that does not bleed or contract to diathermy = devitalised, go higher.
- Severe sepsis: guillotine amputation, complete later. Suction drain for major amputation; avoid stump bandaging (skin breakdown).

Levels

- Ray: metatarsal neck, tennis-racquet incision. Transmetatarsal: plantar flap, excellent walking.
- BKA: long posterior flap (Burgess); tibia cut 14 cm below knee joint (10-12 cm below tuberosity); minimum 7.5 cm. Fibula 2 cm above tibia. Skew flaps if posterior flap skin is not viable.
- AKA: mid-femur, at least 15 cm above tibial plateau (lever arm + room for prosthetic knee); shortest 8 cm below inferior pubic ramus; equal fishmouth flaps.
- Hip disarticulation: high mortality, poor ambulation.

After surgery

- Early: DVT, flap necrosis, infection, pain, haematoma, flexion contracture, psychological. Late: bulbous stump, bone erosion, neuroma, osteomyelitis, ulcer.
- Post-amputation pain about 80%, phantom pain about 70%; no proven treatment. Epidural = best peri-operative analgesia but does not prevent phantom pain. Refer to pain team if first-line oral agents fail.
- Rehab starts immediately: physiotherapy prevents flexion contractures; stump socks shrink the healed stump; PPAM aid (40 mmHg) for early walking; limb fitting at about 6 weeks.

Source: Principles of Limb Ablation and Supportive Therapy [082]

PART 1: LECTURE

Diabetic foot [083] (lecture and LGW)

Topic: Vascular presentations

Burden and causes

- 20% of diabetes admissions are for foot problems; about 15% of diabetics get a foot ulcer each year; 85% of diabetic amputations follow an ulcer. Ulcers are precipitated by trauma.
- Two main causes: neuropathy and ischaemia. Neuropathic 45-60%, ischaemic about 10%, neuro-ischaemic 25-45%; infection is the common final presentation.
- Sensory: injury not felt (tight shoes). Motor: weak short flexors vs long extensors = claw toes, prominent metatarsal heads, ulcer there. Autonomic: less sweating, dry cracked skin.
- Diabetes gives 4-fold PAD risk; diabetes + PAD = highest amputation risk. Microvascular: thick basement membrane, leaky capillaries, sticky platelets.
- Neuropathic ulcer: at pressure points, with a ring of callus.

Wagner grades

- 0 intact skin (impending); 1 superficial; 2 deep to tendon, bone or ligament; 3 osteomyelitis; 4 gangrene of toes/forefoot; 5 gangrene of whole foot.

Acute infected foot

- First control the diabetes (sliding scale, correct ketoacidosis).
- Cultures + swabs, then IV antibiotics covering Staph aureus, strep, pseudomonas, anaerobes. Surface swabs may miss the organism; aspirate tissue fluid for deep culture.
- Tenderness or crepitus = extensive infection. Toe infection spreads to the web space, showing only as mild dorsal redness and tenderness.
- Plain X-ray: osteomyelitis or gas. Drain collections and remove dead tissue within 24 hours.
- Then assess nerves and vessels: absent pulses + low ankle pressure = high-risk foot. Duplex, then CT angiogram; revascularise (angioplasty or surgery) unless healing.

Wound care and prevention

- Do not suture: heal by secondary intention (sutures cut skin edge flow and trap infection). Elevate, avoid oedema.
- Negative pressure therapy: removes fluid, boosts microcirculation, shrinks wound. Total contact cast (with window) for early walking; must be expertly fitted.
- Ulcer care: off-loading, debridement, dressings, antibiotics, revascularisation, amputation; also larvae therapy and BMT. Needs MDT (ortho, vascular, endocrine, podiatry, ID).
- Screen with 10 g monofilament, pulse palpation, ankle pressure.

Charcot foot

- Neuropathy-driven bone fragmentation and joint destruction; medial arch collapse = rocker-bottom foot.

Source: Diabetic Foot lecture [083]; Diabetic Foot LGW [083]

PART 1: LECTURE

Anatomy recap: vascular lower limb

Topic: Vascular presentations

Arteries: hip and thigh

- Aorta, common iliac, then external iliac (lower limb) and internal iliac (pelvis; gives superior and inferior gluteal).
- Femoral artery starts behind the inguinal ligament at the mid-inguinal point, runs through the femoral triangle and adductor canal, then becomes popliteal.
- Branches: superficial circumflex iliac, superficial epigastric, superficial external pudendal; deep external pudendal, profunda femoris, descending genicular.
- Profunda femoris: largest branch, main thigh supply; gives lateral and medial circumflex femoral and 4 perforating arteries.
- Femoral head: mainly retinacular arteries from the circumflex femorals (especially medial); minor supply from artery in ligament of head (obturator).

Anastomoses

- Cruciate (back of thigh, below greater trochanter): transverse limb = transverse branches of medial and lateral circumflex; vertical limb = ascending medial circumflex + descending inferior gluteal above, first perforating below. Links internal iliac and profunda.
- Around ASIS: superior gluteal, ascending lateral circumflex, deep circumflex iliac, superficial circumflex iliac.

Clinical points

- Bleeding control: press femoral artery against the superior pubic ramus, midway between ASIS and pubic symphysis.
- Femoral artery catheter in the femoral triangle for left heart or renal angiography.

Leg and foot

- Popliteal artery: from adductor magnus opening to lower border of popliteus, then splits into anterior (smaller) and posterior (larger) tibial arteries.
- Anterior tibial ends midway between malleoli as dorsalis pedis: superficial, EHL tendon medial, EDL and nerve lateral; enters sole at 1st interosseous space.
- Posterior tibial ends under the flexor retinaculum as medial and lateral plantar. Order front to back: tibialis posterior, FDL, vessels, nerve, FHL ("Tom, Dick, Very Nice Harry").

Veins

- Superficial veins start at the dorsal venous arch; deep veins: femoral and popliteal.
- Great saphenous: in front of medial malleolus, up medial leg and thigh, through cribriform fascia to femoral vein 4 cm below and lateral to pubic tubercle; 15-20 valves.
- Small saphenous: behind lateral malleolus with sural nerve, between gastrocnemius heads; 5-10 valves.
- GSV uses: venous cut-down in front of medial malleolus; reversed graft for coronary bypass (valves). Varicose veins: perforator valve incompetence, deep to superficial flow.

Source: Anatomy Recap ICC1 Vascular Lower Limb

PART 2: TOPIC SUMMARY

Principles of arterial and venous diseases

Acute versus chronic arterial ischaemia

	Acute limb ischaemia	Chronic limb ischaemia
Time course	Under 2 weeks, sudden	Over 2 weeks, gradual
Commonest cause	Embolus (AF, 80% cardiac) or thrombosis on plaque	Atherosclerosis (Buerger's if young smoker)
Key symptom	6 Ps; paralysis = most ominous	Claudication, then rest pain at night, then tissue loss
First steps	IV heparin + analgesia, then revascularise	Stop smoking, exercise, statin, antiplatelet
Procedure	Fogarty embolectomy (embolus); thrombolysis or bypass (thrombosis)	Angioplasty/stent or bypass for disabling claudication, rest pain, ulcer

- Claudication pain is one level below the block: SFA (commonest site) = calf; aortoiliac = buttock and thigh (Leriche: + impotence, absent femoral pulses).
- ABI: 1.0-1.2 normal; 0.5-0.9 claudication; 0.1-0.4 critical ischaemia; over 1.2 = calcified vessels (diabetes), unreliable.
- After revascularisation watch for reperfusion injury (hyperkalaemia, acute tubular necrosis) and compartment syndrome (pain on passive stretch, pulses may persist): urgent fasciotomy.
- Trauma: hard signs (absent pulse, active bleeding, expanding haematoma, bruit) = straight to theatre; soft signs = ABI, arteriography if under 0.9. Knee dislocation + pulseless foot = popliteal injury.

Venous disease

- DVT: Virchow's triad. Wells under 2 = D-dimer first; over 2 = duplex directly. Treat LMWH with warfarin or a NOAC; IVC filter only if anticoagulation is contraindicated.
- Varicose veins: great saphenous = medial; small saphenous = posterolateral calf. Duplex before treatment. Trendelenburg test controls the SFJ.
- Treatment ladder: stockings, sclerotherapy, endovenous laser, SFJ ligation and stripping.

Ulcer comparison

	Venous	Arterial	Neuropathic
Site	Gaiter area, medial malleolus	Toes, lateral malleolus	Under metatarsal heads
Look	Shallow, sloping edges, pigmentation	Punched out, no granulation	Callus ring
Pain	Mild	Severe	Painless
Key treatment	Compression (only if ABPI over 0.8)	Revascularise	Off-load pressure

PART 2: TOPIC SUMMARY

Aneurysms

Definitions

- Aneurysm = permanent localised dilatation to 1.5 times normal diameter. True = all three layers; false (pseudoaneurysm) = haematoma communicating with the lumen, usually after femoral puncture.
- AAA is the commonest; about 95% are infrarenal. Smoking is the key risk factor.

Abdominal aortic aneurysm

- Usually found by chance on ultrasound; may cause a pulsatile mass or back pain.
- Rupture: abdominal + back pain + hypotension; mortality over 80%; can mimic ureteric colic, pancreatitis or MI. Treat with ABCDE, permissive hypotension, CTA if stable, immediate repair.
- Ultrasound = screening and surveillance; CTA = planning repair. Screen all men from 65.
- Repair asymptomatic AAA at 5.5 cm. EVAR has lower early mortality than open repair but needs lifelong surveillance (endoleak, migration).
- Late open-repair danger: aorto-enteric fistula (GI bleed).

Other aneurysms

Site	Key facts	Repair when
Popliteal	Commonest peripheral; 50% bilateral; check the aorta; threatens the limb (thrombosis, emboli)	Over 2 cm
Femoral	Second commonest peripheral	Over 2.5 cm
Splenic	Commonest visceral; young women, pregnancy	Symptoms, growth, over 2.5 cm, pregnancy
Renal	Hypertension	Hard-to-control BP or over 2 cm
Femoral pseudoaneurysm	After puncture	Over 2.5 cm: ultrasound-guided thrombin injection

PART 2: TOPIC SUMMARY

Vascular presentations

History and examination

- Claudication distance and site; rest pain eased by hanging the leg down; post-meal pain with weight loss = mesenteric ischaemia; TIA or amaurosis fugax = carotid disease.
- Pulses from proximal to distal: femoral at the mid-inguinal point, popliteal deep with knee flexed, posterior tibial behind medial malleolus, dorsalis pedis lateral to EHL (absent in 10%).
- Buerger's angle under 20 degrees = severe ischaemia. Capillary refill over 5 s = significant ischaemia.
- Venous examination standing; Trendelenburg test for SFJ incompetence; modified Perthes' test for deep vein occlusion.

Carotid, renal and mesenteric disease

- Carotid: duplex first; symptomatic significant stenosis = endarterectomy within 2 weeks. CEA risks: stroke 2-3%, cranial nerves XII, X, IX.
- Dialysis access: native fistula made 6 months before need; rule of 6s for a mature fistula.
- Acute mesenteric ischaemia: pain out of proportion to signs, raised lactate; CTA; revascularise and resect dead bowel. Chronic: pain 30 min after meals, food fear, weight loss.

Chronic limb swelling

- Unilateral = local (DVT, cellulitis, lymphoedema); bilateral = systemic (heart, kidney, liver failure, drugs such as CCBs).
- Lymphoedema: Stemmer's sign; primary praecox = commonest (girls after menarche); secondary = cancer, surgery, radiotherapy, filariasis. Treat with skin care, massage, compression.
- Post-thrombotic syndrome: heaviness, pigmentation, ulcer after DVT; lifelong compression.

Diabetic foot and amputation

- Neuropathy + ischaemia + infection. Wagner 0 intact, 1 superficial, 2 deep, 3 osteomyelitis, 4 forefoot gangrene, 5 whole foot.
- Infected foot: control glucose, cultures, IV broad antibiotics, drain within 24 h, then assess pulses and revascularise. Never suture: heal by secondary intention.
- Amputation: revascularise first if possible. BKA tibia cut about 14 cm below the knee joint (long posterior flap); AKA at least 15 cm above the tibial plateau. Physiotherapy prevents flexion contracture; phantom pain about 70%.

Anatomy anchors

- Femoral artery at the mid-inguinal point; profunda femoris = main thigh supply and key collateral; cruciate anastomosis links internal iliac and profunda.
- Great saphenous vein joins the femoral vein 4 cm below and lateral to the pubic tubercle.